Volcanoes: Windows into a Restless Planet

A volcano is an opening in a planet's crust through which molten rock, ash and gases escape to the surface. The molten rock is called magma while it is underground and lava once it has erupted. Volcanoes build up over time as layers of cooled lava and fragments of rock pile up around the vent.

Most volcanoes form near the edges of tectonic plates, the large slabs that make up Earth's outer shell. At subduction zones, one plate sinks beneath another and releases water that helps the rock above it melt. At spreading ridges, plates move apart and magma rises to fill the gap, which creates new crust on the ocean floor.

Some volcanoes sit far from any plate boundary. These are usually linked to hotspots, long-lived plumes of unusually hot rock rising from deep inside the planet. As a plate drifts slowly over a hotspot, a chain of volcanic islands can form, with the youngest island sitting above the hotspot and the oldest ones carried farthest away. The Hawaiian Islands are a well-known example.

Volcanoes come in several shapes. Shield volcanoes have broad, gentle slopes built from runny lava that spreads over long distances. Stratovolcanoes, also called composite volcanoes, are steep and cone-shaped, and are made of alternating layers of lava and ash. Cinder cones are small, steep hills formed from fragments of lava thrown into the air.

The stickiness of magma, known as its viscosity, strongly affects how an eruption behaves. Runny magma lets gas escape easily, so it tends to flow out quietly as lava streams. Thick, sticky magma traps gas, and the pressure can build until the volcano blows apart in a violent explosion.

Eruptions release much more than lava. Ash is made of tiny fragments of rock and glass that can be carried far downwind and can damage aircraft engines. Pyroclastic flows are fast, hot avalanches of gas and volcanic debris that race down a volcano's slopes. Volcanoes also emit gases such as water vapour, carbon dioxide and sulfur dioxide.

Lava itself comes in different textures. Pahoehoe lava has a smooth, ropy surface, while aa lava is rough and covered in jagged clinkers. Both words come from the Hawaiian language. When lava flows into the sea or erupts under water, it cools quickly and can form rounded blobs known as pillow lava.

A caldera is a large bowl-shaped depression that forms when a volcano empties its magma chamber and the ground above collapses. Calderas can be many kilometres across, and some later fill with water to become lakes. Crater Lake in Oregon sits in a caldera that formed after a huge eruption of an old volcano.

Volcanologists classify volcanoes by how recently they have erupted. An active volcano has erupted in recent history or is showing signs of unrest. A dormant volcano is quiet at present but could erupt again, while an extinct volcano is not expected to erupt any more. These labels are not always certain, because some volcanoes have woken after very long periods of silence.

Scientists try to forecast eruptions by watching for warning signs. Small earthquakes often increase as magma moves upward, the ground may swell as pressure builds, and the amount and type of gas escaping from vents can change. Combining several of these measurements gives a better warning than relying on any single one.

Volcanic activity has benefits as well as dangers. Weathered volcanic ash and lava often produce rich, fertile soils that farmers value. Hot rock near the surface supplies geothermal energy that is used for heating and for generating electricity. Volcanic activity also created many of the islands and much of the land that people live on today.

Large eruptions can influence the climate. When a powerful eruption sends sulfur dioxide high into the atmosphere, it forms tiny droplets that reflect sunlight back into space. This can cool the planet slightly for a year or two. The effect fades as the droplets gradually fall out of the air.

Volcanoes are not unique to Earth. Olympus Mons on Mars is a giant shield volcano and one of the largest known volcanoes in the solar system. Jupiter's moon Io is the most volcanically active body known, with eruptions that throw sulfur-rich material high above its surface.

People have long lived close to volcanoes despite the risks, drawn by the good soil and the beauty of the land. Communities near active volcanoes often prepare with hazard maps, warning systems and evacuation plans. Education also matters, because residents who know the warning signs and escape routes are far better protected when an eruption begins.